Maryam Arif (Links attached to images)

Islamabad, Pakistan | ranamaryamarif@gmail.com | Press Coverage | My Website | LinkedIn | Medium | **Portfolio**

ACHIEVEMENTS

1. **Massachusetts Institute of Technology (MIT)** Competing in research-driven challenges on data science and decision-making. [Link]
2. **Harvard University:** Selected 3 times for Harvard University's largest annual student conferences, featuring speakers from Nobel Prize winners to heads of state.
3. **University of Cambridge Judge Business School:** Reviewed 30+ research papers to identify gaps in the current understanding of social constraints in female entrepreneurship [Link]
4. **Presidential Award:** Awarded for excellence in research and community impact. [Link]
5. **Youngest National Author:** Authored 10+ books at the age of 16 on data-driven policy, society and various other topics, featured by the U.S. Embassy, Pakistan Embassy, National television Channels & Newspapers.
6. **TEDx**: *Guest Speaker*, discussed how language can overcome socio-cultural barriers, focusing on its role in shaping educational access and identity.
7. **Research Impact Achievement:** Achieved significant recognition for my research contributions, with my work reaching a substantial number of academic reads, reflecting the impact and relevance of my findings in the field**.**
8. **National Youth Award**: Issued by the Government of Pakistan for contributions to research and data analytics. [Link]
9. **Asian Excellence Awards:** Led research operations with strong analytical and financial-engineering oriented methodologies, highlighting education in South Asia.
10. **International Olympiad on World Economy**: First Place & Honors Diploma against 33 international teams [Link]
11. **Business in Trade Directory:** Presented at the President's House (May 2022), showcasing technology solutions aimed at bridging gaps in business, research operations and financial engineering.
12. **100 Young Influencers of Pakistan**: Advocated for female micro-entrepreneur empowerment, advancing research in decision-making. Analyzed 500+ data points, with the project reaching 8 million views [Link]

EDUCATION

COMSATS University Islamabad (University website) **Pakistan**

Bachelor of Business Administration (CGPA: 3.28/4.0) transcript **Sep 2021 – Sep 2025** *(4 years degree)*

- **Relevant coursework:** Research tools and techniques, Production and Operations Management, Financial Management, Strategic Management, Statistical inference, Business Finance, Business Mathematics.
- **Youngest in my batch:** Admitted to university at the age of 16 (started at 17); completed 139 credits (exceeding the 120-credit requirement) by taking additional courses.

International Exchanges:

- **China, Beijing Language and Culture University (2025):** Enhanced cross-cultural and analytical skills. (Details)
- **USA, University of Oregon and Portland State University (Oregon, 2024)**: Selected from 125+ entrepreneurs at University of Oregon and Portland State University (2024) to refine decision-making and engage with leaders, including Congresswoman Val Hoyle and Eugene Mayor Lucy Vinis (Details)

PUBLICATIONS

Research and Writings: (* = *First Author and correspondent)*

1. **Maryam Arif** * & Saeed, S. (2025). ML-Driven Quantum Portfolio Optimization: Hybrid CNN-LSTM Architecture with Adaptive Zero-Noise Extrapolation on NISQ Devices [Pre-Print]
2. **Maryam Arif** * & Saeed, S. (2025). Machine learning predictive analytics for social media-enabled women's economic empowerment in Pakistan (arXiv:2512.12685). [Pre-Print] [Data & Code] [Conference Poster]
3. **Maryam Arif,** (2025). Language-in-Education Policies in Pakistan: Evidence for an Equity-Driven Multilingual Framework for Urdu, English and Regional Languages. [Pre-Print] [Conference Poster]
4. **Maryam Arif** * & Saeed, S. (2025). Addressing Educational Inequities and Marginalization in Pakistan: A Comparative Analysis of Policy and Practice. [Pre-Print] [Data & Code] [Conference Poster]
5. **Maryam Arif**, (2022). Entrepreneurial ecosystems in emerging economies [Research Paper] [Presentation Slides]
6. **Maryam Arif**, Invisible chains: The impact of social constraints on female microentrepreneurs' supplier dependency and business performance. **Research with the University of Cambridge, UK** [Collaborative Research].
7. **Maryam Arif**. (2020). Progression and microenvironment of cancer by lactic acid. Pakistan Journal of Pharmacology, 34(1), 23–29. (Blind peer-to-peer review). **Published at the age of 15** [Research Paper]

Books:

Published 10+ books by 16, nationally recognized for work in problem-solving, decision-making, and cognitive development.

1. **Inspiring and Keep Calm Quotes: Coloring Book for Adults**
   Uses motivational frameworks to support emotional regulation, illustrating early engagement with behavioral decision-making and stress-optimization techniques. ASIN: B088N519CM, [Link]
2. **Face Your Fears: Master Anxiety, Fear and Emotions. Build Your Self-Confidence and Self-Esteem**
   Explores structured approaches to overcoming cognitive barriers paralleling concepts of risk assessment and resilience under uncertainty. ASIN: B087ZYKTQ1, [Link]
3. **Reasons and Morals**
   Analyzes ethical reasoning and structured judgment, reflecting principles of rational decision-making and strategic evaluation. ASIN: B087PJ9GS1, [Link]
4. **Coloring Book for Children: For Kids (4–8)**
   Supports early cognitive development and pattern recognition, foundational to logical thinking and problem-solving. ASIN: B088VQ4FKB, [Link]
5. **Travelling to Unicorn Island**
   Encourages imagination and narrative reasoning, cultivating early skills in structured thinking and creative problem formulation. ASIN: B088MM9JW6, [Link]
6. **Unlimited Power: The Secret of Achieving More**
   Discusses goal setting and performance strategies analogous to optimization, efficiency, and systematic improvement. ASIN: B087NMK2GH, [Link]
7. **Cakes and Cupcakes: Creative Haven Coloring Book for Adults**
   Cultivates pattern recognition, attention to detail and structured creativity skills applicable in modeling and analytical visualization. ASIN: B088N519CM, [Link]
8. **The Book of Ideas: Journal**
   Promotes systematic reflection, iterative thinking, and structured ideation core components of mathematical modeling and research. ASIN: B08FKH6P5N, [Link]
9. **The Cupcake Party**
   Encourages narrative structure and logical sequencing, reflecting early interest in organized thinking and design processes. ASIN: B08FKH6P5N, [Link]
10. **Zen and Anti-Stress: Coloring Book for Adults**
    Integrates mindfulness with structured expression, echoing concepts of optimization under stress and cognitive clarity in decision processes. ASIN: B088N519CM, [Link]

Additional Publications: For a comprehensive list of books and further details, please visit my Author Profile, as well as the newspaper and television features on my work [Link]

RESEARCH EXPERIENCE

Federation of Pakistan Chambers of Commerce — Islamabad, Pakistan
Committee Member, Researcher — *Apr 2022 - Present*

- Analyzed and evaluated educational initiatives aimed at enhancing women's literacy in financial and business-related linguistics, empowering over 3,000 women with communication skills essential for economic participation.

University of Cambridge, Judge Business School — Remote, United Kingdom
Research Assistant — *Aug 2024 - May 2025*

- Contributed to high-impact research on female microentrepreneurs, conducting literature reviews, data analysis, and writing to support evidence-based strategies for business performance improvement
- Advisor: Soniya Gupta-Rawal, Jaideep Prabhu

COMSATS University Islamabad — Islamabad, Pakistan
Student Researcher — *May 2025 - Oct 2025*

- Researched on projects related to international relations and strategic management, contributing data analysis and literature reviews.
- Advisor: Uzma Naeem

INDUSTRY EXPERIENCE

Total Years of Industry Experience = 5 years 1 month, worked alongside university.

Innovista Private Limited, **Research Consultant** — **May 2025 – Sep 2025 (5 mos)**

- Research insights into linguistic and cultural aspects of national anthems, analyzing phonetics, tone, and rhythm for cross-cultural understanding.

MABEAUTY, **CEO and Research head** *(remote)* — **2021 – Present (4 years)**

- Small business: Research and development of beauty products, ensuring each formula is 100% organic with natural ingredients sourced from ethical suppliers.

Research and Innovation Program — **Jun 2023 – Jan 2024 (8 mos)**

- Conducted research on innovative entrepreneurship models in Pakistan, contributing to the development of business incubation programs